

Roll No. _____

B.Sc. (Hons.) SEMESTER V EXAMINATION 2016-17**PHYSICS****BPT - 501 : Mathematical Physics****Time : Three hours****Max. Marks : 70****(WRITE YOUR ROLL NO. AT THE TOP IMMEDIATELY ON THE RECEIPT OF THIS QUESTION PAPER)****Note:** Answer all questions. The figure in the right hand margin indicates marks. Unless otherwise specified the mathematical symbols have their usual meanings.**1. Answer any five of the following :****5 × 4 = 20**

- Find the magnetic induction $\vec{B}$ (in spherical polar coordinates), given the magnetic vector potential $\vec{A} = \frac{\mu_0}{4\pi} \frac{\vec{M} \times \vec{r}}{r^3}$. Here $\vec{M} = M\hat{z}$, i.e., a vector of constant magnitude pointing in the z direction.
- What are covariant and contravariant vectors? Explain with a physics example each.
- If A is a (nonsingular) symmetric matrix, show that A^{-1} is also symmetric.
- Evaluate $\vec{\nabla} \cdot \frac{\vec{r}}{r^3}$. What happens at the origin? Explain.
- Find the Fourier series of the function $\sin^2(x)$, $(-\pi \leq x \leq \pi)$.
- Normalize the Laguerre polynomial, $L_2(x) = 1 - 2x + \frac{x^2}{2}$. Construct a second order, ordinary, linear, homogeneous differential equation (2-ODE) that is satisfied by this polynomial.

— (2) —**2. (a) Verify the following (in $\mathbb{R}^3$):**

$$\begin{aligned}\varepsilon_{ijk}\varepsilon_{ipk} &= 2\delta_{jp} \\ \varepsilon_{ijk}\varepsilon_{pqk} &= \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}\end{aligned}$$

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- Write down the spherical polar unit vectors in terms of Cartesian unit vectors. Using these calculate the partial derivatives of $\hat{r}, \hat{\theta}, \hat{\phi}$ with respect to r, θ, ϕ .

6 $\frac{1}{2}$ **(OR)**

- Determine the circular cylindrical coordinate components of velocity and acceleration of a moving particle with position vector $\vec{r}(t)$.
- Given a vector in spherical coordinates: $\vec{P} = \frac{2p \cos \theta}{r^3} \hat{r} + \frac{p \sin \theta}{r^3} \hat{\theta}$, express it in Cartesian coordinates.

6**6 $\frac{1}{2}$**

3. (a) $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin(x)$ is one solution of Bessel equation. Find a linearly independent second solution (up to a constant). **5**
- (b) Given the generating function $g(x, t) = e^{-x^2} e^{2xt - t^2} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}$, show that $\int_{-\infty}^{\infty} e^{-x^2} [H_n(x)]^2 dx = 2^n n! \sqrt{\pi}$. **7½**

(OR)

- (a) Using the generating function $g(x, t) = \frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n$, ($|t| < 1$), for Legendre Polynomials, show that

$$P'_{n+1}(x) - P'_{n-1}(x) = (2n+1)P_n(x).$$

(Here ' stands for derivative with respect to x). **5**

- (b) Solve $(1-x^2)y'' - 2xy' + n(n+1)y = 0$, using the Frobenius method (about $x = 0$). Show that for integral values of n one of the solutions results in a polynomial. **7½**

4. (a) By expressing the vector cross products in terms of the Levi-Civita symbols (ε_{ijk}), verify that $(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C})$. **5**
- (b) Find the eigenvalues and (normalized) eigenvectors of A :

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \sqrt{2} \\ 0 & \sqrt{2} & 0 \end{pmatrix}$$

How are these related to the eigenvalues and eigenvectors of A^{-1} ? **7½**

5. (a) Sketch a periodic square wave function that is also an even function. Choose the period to be 2, and amplitude to be 1 (for half the period it is 0 and for the other half it is 1). Find the Fourier series for this periodic function. **6½**
- (b) The transformation relation between the unit vectors in circular cylindrical coordinates and Cartesian coordinates can be written as:

$$\begin{pmatrix} \hat{\rho} \\ \hat{\theta} \\ \hat{z} \end{pmatrix} = C \begin{pmatrix} \hat{i} \\ \hat{j} \\ \hat{k} \end{pmatrix}$$

Write down the matrix C and show that it is orthogonal. **6**

Useful information: $\int_0^{\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$ (n is an integer)

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External Portals & Academic Resources:

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app>